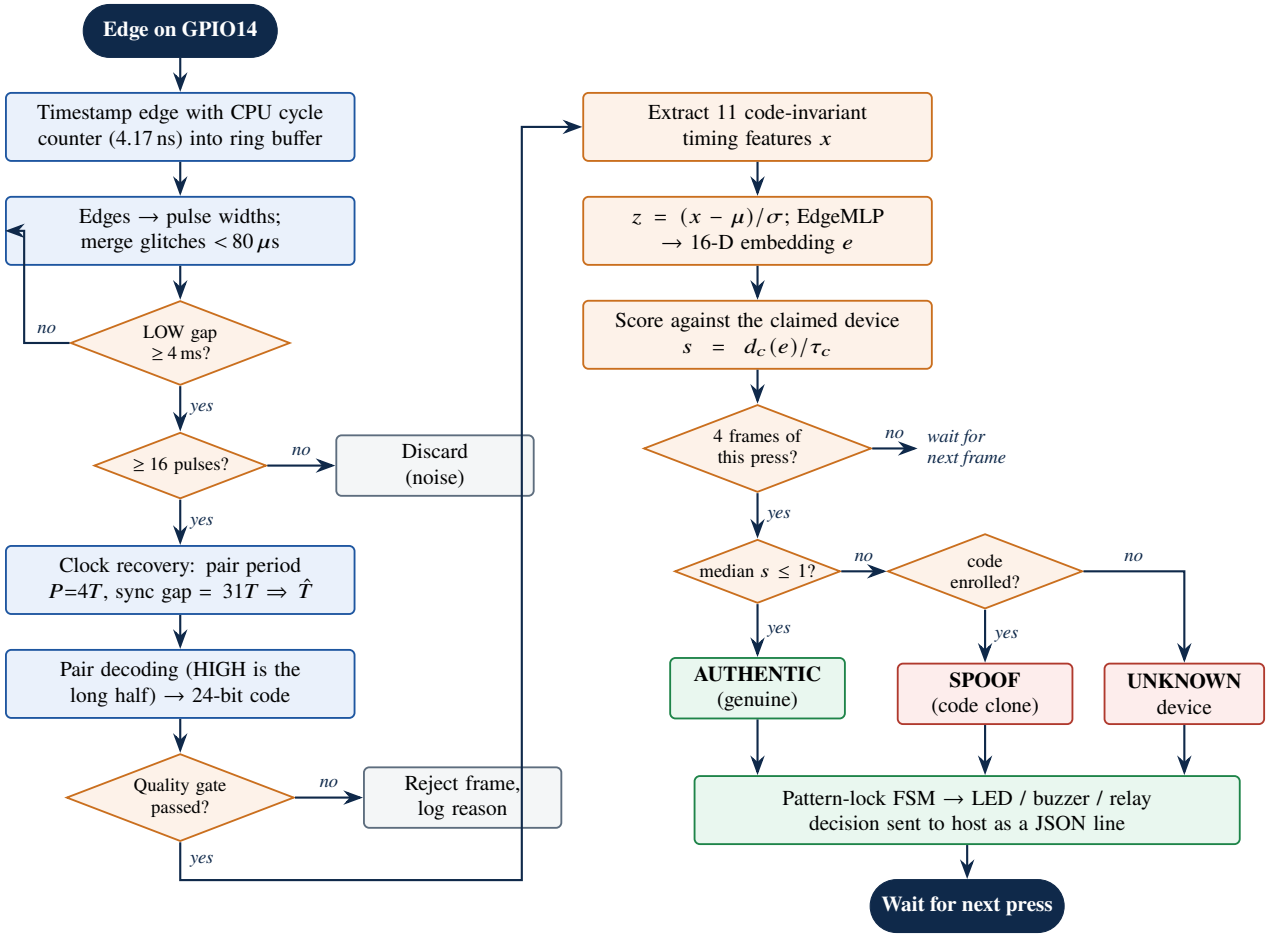

Edge on GPIO14
Timestamp edge with CPU cycle counter (4.17 ns) into ring buffer
Edges → pulse widths; merge glitches < 80 µs
LOW gap ≥ 4 ms?
no
yes
≥ 16 pulses?
no
Discard (noise)
yes
Clock recovery: pair period P=4T, sync gap = 31T ⇒ T̂
Pair decoding (HIGH is the long half) → 24-bit code
Quality gate passed?
no
Reject frame, log reason
yes
Extract 11 code-invariant timing features x
z = (x − µ)/σ; EdgeMLP → 16-D embedding e
Score against the claimed device s = d_c(e)/τ_c
4 frames of this press?
no
wait for next frame
yes
median s ≤ 1?
no
code enrolled?
no
yes
yes
AUTHENTIC (genuine)
SPOOF (code clone)
UNKNOWN device
Pattern-lock FSM → LED / buzzer / relay decision sent to host as a JSON line
Wait for next press